

Final Project Report: Build the Algorithmic Black Box (WiDS)

1. Introduction and Motivation

Financial markets are intricate systems where prices are determined by the interaction of many agents placing buy and sell orders. Contemporary electronic markets function on the principle of limit order books, where orders are matched according to price-time priority. It is critical to understand this market microstructure to effectively design and analyze algorithmic trading strategies, particularly in the context of data-driven and learning-based approaches.

The objective of this project, undertaken as a part of the WiDS initiative, was to incrementally develop an “algorithmic black box” – a simplified yet realistic market simulator that captures the fundamental mechanics of an electronic market. Instead of concentrating on the profitability of trading, the importance of creating a correct, modular, and extensible simulation framework was highlighted. This is a necessary precursor to any meaningful experimentation with reinforcement learning or adaptive trading strategies.

During the course of the WiDS initiative, the project progressed from learning about fundamental concepts of order books to developing a fully functional event-driven simulator with agents, analytics, and visualization. This report chronicles the learning process, design choices, experiments, and findings made during the course of the program.

2. System Overview

The final system is made up of a number of interlocking parts, each of which is intended to simulate an exchange architecture in a simplified way. At the heart of the system is the limit order book and matching engine, which determine how orders are stored and how trades are made. Around this core is the simulation engine, which controls the passage of time and the processing of events. On top of the market infrastructure are the trading agents, which produce order flow and interact with the market. Finally, there are analytics and visualization components that watch the state of the market and produce numerical and graphical results.

This modular structure allows for a clean separation of concerns. The mechanics of the market are decoupled from the trading agents, so that trading strategies can be changed or replaced without changing the market. Similarly, the analytics are passive observers of the market.

3. Methodology

3.1 Limit Order Book Design

The market structure is modeled using a limit order book with separate bid and offer sides. Orders are grouped by price, and for each price, a first-in-first-out (FIFO) queue is kept. This ensures that the price-time priority rule is maintained, which is a basic tenet of how actual exchanges are organized to treat all traders fairly.

The book is organized to handle limit orders and market orders. A limit order contains both price and size and can either be posted to the book or executed immediately if it crosses the spread. A market order contains only size and is executed at the best available prices until the order is completely filled or the market runs dry. The order book model has given valuable insights into how market liquidity is organized and how even a small flow of orders can affect prices and spreads.

3.2 Matching Engine

The matching engine is tasked with the responsibility of executing trades whenever there are matching buy and sell orders. Buy orders are matched against the lowest available ask prices, while sell orders are matched against the highest available bid prices. Partial fills are allowed, and the orders are deleted from the book once they are fully executed. Each trade is recorded in a structured trade tape that includes the execution time, side, price, and quantity. The trade tape provides the foundation for all other analytics. The development of the matching engine further emphasized the need for deterministic logic in market simulations.

3.3 Event-Driven Simulation Engine

The simulation operates in discrete time using an event-driven architecture. Events are processed chronologically and include order submissions, market order executions, and snapshot events that capture the market state at regular intervals.

This approach avoids reliance on real-time delays and allows the simulation to be reproducible and deterministic. It also makes the system easier to debug and extend, as all state transitions occur through clearly defined events.

3.4 Agent Interaction

To generate order flow, a simple random trading agent is introduced. At each simulation step, the agent randomly chooses between buy, sell, or hold actions and submits small market orders when feasible. While this agent does not aim to be profitable, it serves an important role in validating the correctness of the market infrastructure.

The use of a deliberately simple agent aligns with the project's emphasis on system design rather than strategy optimisation. It ensures that observed market behaviour arises from the exchange mechanics rather than complex agent logic.

4. Market Analytics and Visualisation

From the recorded trades and snapshots, several standard market microstructure metrics are computed:

- **Bid-Ask Spread:** The difference between the best ask and best bid prices, serving as a proxy for market liquidity.
- **Mid-Price:** The average of the best bid and best ask prices, often used as a reference price.
- **VWAP (Volume-Weighted Average Price):** A weighted average of execution prices, reflecting the average price at which trades occurred.
- **Mid-Price Volatility:** The standard deviation of mid-price changes over time.

Snapshot data is further aggregated into OHLC (open, high, low, close) bars, enabling the construction of candlestick charts. These visualisations provide intuitive insight into price stability and variability during the simulation.

5. Experiments and Observations

A baseline simulation is conducted with seeded initial liquidity on both sides of the order book. The random agent generates market orders that are processed by the matching engine.

The observed market behaviour is stable:

- The bid–ask spread remains constant due to the absence of price-improving limit orders.
- The mid-price shows minimal variation, reflecting a balanced and liquid market.
- VWAP aligns closely with the mid-price, as expected in a stable environment.
- Volatility remains low, indicating limited price discovery.

While these results are simple, they confirm that the simulator behaves consistently and that the analytics correctly reflect the underlying market state.

6. Learnings from the WiDS Program

Throughout the WiDS program, this project provided hands-on exposure to several key concepts:

- The mechanics of limit order books and price–time priority.
- The importance of event-driven architectures for financial simulations.
- The challenges of maintaining consistent state updates in dynamic systems.
- The distinction between building market infrastructure and designing trading strategies.

An important learning was that reliable market simulation is a prerequisite for any meaningful application of reinforcement learning. Attempting to train agents without a correct and stable environment would lead to misleading results.

7. Limitations and Future Work

The current implementation focuses on infrastructure correctness rather than trading performance. As a result, the agent behaviour is intentionally simple and does not adapt to market conditions. The simulation also omits transaction costs, latency, order cancellations, and strategic limit order placement.

Future extensions could include:

- Integration of reinforcement learning agents.
- Inventory-aware reward functions.
- Multiple interacting agent populations.
- More realistic market frictions such as latency and fees.

The modular design of the system makes these extensions feasible without major architectural changes.

8. Conclusion

This project successfully delivers a complete, end-to-end market simulation framework developed over the course of the WiDS program. By prioritising correctness, modularity, and clarity, the system provides a strong foundation for future experimentation with learning-based trading strategies. While the current agents are simple, the insights gained from building the exchange infrastructure are substantial and align closely with the learning objectives of WiDS.